

THE CLOSED CAUSAL LOOP & EPISTEMIC BOUNDARY SYNTHESIS

Synthesizing All 17 Chapters Across 4 Parts: Sensory Transduction, Cognitive Deliberation, Real-Time Invariant Enforcement, and Physical Dynamics

PART I: FOUNDATIONS (Ch 1–4)

Body, Brain & Nervous System

Ch 1: The Causal Boundary
Ch 2: Physical Body & Dynamics
Ch 3: Learned Brain & Latency
Ch 4: Nervous System & Clocks

PART II: DATA & LEARNING (Ch 5–7)

Demonstrations, Training & Bounds

Ch 5: Sensor Demonstration Data
Ch 6: Imitation & RL Training
Ch 7: Evaluation & Rare Events
Finite-Sample Exposure Bounds

PART III: ONLINE RUNTIME (Ch 8–13)

Perception to Edge Placement

Ch 8: Perception · Ch 9: Memory
Ch 10: Intent · Ch 11: Planning
Ch 12: Real-Time Safety Filter
Ch 13: Heterogeneous Placement

PART IV: GOVERNANCE (Ch 14–17)

Verification, Release & Frontier

Ch 14: Shared Intervention
Ch 15: Empirical Stress Testing
Ch 16: Formal Safety Release
Ch 17: The Epistemic Frontier

THE BRAIN: COGNITIVE DELIBERATION

High-Capacity Stochastic Policy (MPU)

Cadence: 20–50 Hz · Linux / NPU · Latency τ_{infer}

- VLM Semantic Goal Grounding (Ch 10)
- SE(3) Dynamic Kinematic Frame Tree (Ch 9)
- Diffusion / ACT Action Chunks $H=16$ (Ch 11)
- Expiring Intent Lease Issuance t_{expire} (Ch 10)
- Epistemic Uncertainty & Silent Drift (Ch 7, 17)

EMITS: Candidate Action Trajectory $\hat{u}_t:t+H$

Sensor Stream y_t (DMA)

SENSORY TRANSDUCTION & INFORMATION FRESHNESS

Photons / Forces → Digital Bits (The Sensing Boundary)

- Transduction: MIPI CSI-2 CMOS, IMU gyros, optical wheel encoders (Ch 8)
- Zero-Copy DMA: Ring buffers, hardware PTP nanosecond timestamps
- Information Age Freshness: $\tau_{\text{total}} = \tau_{\text{readout}} + \tau_{\text{DMA}} + \tau_{\text{infer}} + \tau_{\text{act}}$
- Spatial Displacement Lag: Unchecked motion $d_{\text{lag}} = v \cdot \tau_{\text{total}}$ (Ch 2, 17)
- Endogenous Feedback: Action u_t shifts pose, shaping future observations y_{t+1}

PROPOSAL-PERMISSION CAUSAL INTERFACE

Hardware Architectural Separation (Ch 13, 17)

Shared SRAM TCM:

Zero-copy lock-free seqlock mailbox

Dead-Man Leases:

Lease t_{expire} bounds MPU stalls ≤ 50 ms

Invariant Projection:

$u_t^* = \text{argmin} \|u - \hat{u}\|^2$ s.t. $h(x) \geq 0$

Authority Hierarchy:

Reflex Stop > Human Override > Policy

Gate

THE NERVOUS SYSTEM: REFLEX ENFORCER

Deterministic Invariant Filter (MCU)

Cadence: 1000 Hz · FreeRTOS / Cortex-M · Hard Real-Time

- Real-Time Active-Set QP Solver: $h(x) \geq 0$ (Ch 12)
- Dynamic Stopping Clearance: $d_{\text{stop}}(v) \leq d_{\text{gap}}$ (Ch 2)
- Lock-Free Inter-Core Mailbox & Lease Check (Ch 4)
- Shared Intervention & C² Bumpless Takeover (Ch 14)
- Hardware Watchdog & ISO 13849 Safety Relay (Ch 15)

PERMITS: Verified Motor Commands u_t^*

Actuation Torques u_t^*

THE PHYSICAL PLANT & NEWTONIAN REALITY ($W_t \rightarrow W_{t+1}$)

Continuous Mechanics, Conservation Laws & Irreversibility

- Inertial Momentum: Kinetic energy $E_k = 1/2 m v^2$ (No software rollback!)
- Tire/Surface Coulomb Friction Limits: $F_{\text{friction}} \leq \mu N$ (Ch 2, 17)
- Non-Smooth Contact Dynamics: Painlevé paradoxes & brittle yield (Ch 2)
- Actuator Dissipation: Thermal Joule heating $P_{\text{loss}} = I^2 R$, winding limits
- Physical State Update: $W_{t+1} = \text{Dynamics}(W_t, u_t^*, \text{Disturbances})$

CHAPTER 17 FRONTIER SYNTHESIS: THE BOUNDARY OF VERIFIABLE KNOWLEDGE

1. Detectorless Blind Spots

States S_A & S_B produce identical telemetry prior to deadline t_{deadline} (internal voids, friction drops, drift).

2. Astronomical Exposure Limits

Proving $p \leq 10^{-9}/h$ takes $n \geq 3.0 \times 10^9 h$ (342k yrs).
Empirical sampling cannot certify open-world safety.

3. Structural Containment Mandate

Where evidence ends, nervous system clamps authority (v_{clamp} , force limits) to render the unknown harmless.